Physics 9th – Chapter-5-Pressure and Deformation in Solids (Short Response Questions)

Q1. While walking on a trampoline, do you feel more pressure when you stand still or jump up and down? Why does pressure change with movement?

Ans:

- We feel more pressure when we jump up and down.
- Standing still, the force on the trampoline is only our weight, so $P = W/A$.
- On landing, the trampoline must also stop our downward motion ($F = \Delta p/\Delta t$), so the force becomes greater than our weight; for the same area the pressure ($P = F/A$) increases.

Q2. How does the shape of a thumb pin help it penetrate surfaces easily?

Ans:

- The pointed tip of a thumb pin has a very small area, so the force of the thumb produces a very large pressure ($P = F/A$) that pierces the surface.
- The broad flat head has a large area, so the pressure on the thumb is small and it does not hurt.
- Concept: for the same force, pressure is inversely proportional to area.

Q3. If you blow up a balloon and then tie it closed, why does it stay inflated even though you stop blowing? How does pressure play a role here?

Ans:

- The air molecules trapped inside keep colliding with the inner walls and exert pressure equally in all directions.
- This inside pressure is slightly greater than atmospheric pressure and is balanced by the elastic (stretching) force of the rubber.
- As the tied balloon does not let air escape, this pressure remains, so the balloon stays inflated.

Q4. Why an inner airtight layer of a space suit is designed to maintain a constant pressure around the astronaut?

Ans:

- Space is almost a vacuum, so there is no outside pressure on the body.
- Without pressure, gases dissolved in the blood would form bubbles, body fluids would boil and the lungs could be damaged.
- The airtight layer keeps a constant pressure (like on Earth) so the astronaut can breathe and the body works normally.

Q5. If a liquid has density twice the density of mercury, what will be height of liquid column in barometer?

Ans:

- Atmospheric pressure $P = \rho gh$ is the same, so h is inversely proportional to density ($h \propto 1/\rho$).
- Mercury column at sea level = 760 mm.
- Height of new liquid = 760 mm ÷ 2 = 380 mm = 38 cm.

Q6. Why we wouldn't be able to sip water with a straw on the Moon?

Ans:

- We drink with a straw by lowering the pressure in the mouth; the atmospheric pressure on the liquid surface then pushes the liquid up the straw.
- The Moon has no atmosphere, so there is no atmospheric pressure to push the water up.
- Hence sucking cannot make water rise in the straw on the Moon.

Q7. How are we able to break a metal wire by bending it repeatedly?

Ans:

- Each bend stretches the wire beyond its elastic limit, causing permanent (plastic) deformation at the bend.
- Repeated bending forms and grows tiny cracks (metal fatigue) and the metal there becomes weak and hot.
- Finally the stress exceeds the breaking point and the wire breaks.

Q8. A spring having spring constant k when loaded with mass m is cut into two equal parts. One of the parts is loaded with the same mass m again. What will be its spring constant now?

Ans:

- Under the same force, each half of the spring stretches only half as much as the whole spring ($x' = x/2$).
- From Hooke's law $k = F/x$, so $k' = F/(x/2) = 2k$.
- So the spring constant of each half becomes double ($2k$).

Q9. Why do static fluids always exert a force perpendicular to the surface?

Ans:

- A fluid at rest cannot support a force parallel to a surface; any parallel component would make the fluid flow.
- Pressure in a static fluid acts equally in all directions (Pascal's principle), so the force $F = PA$ acts at right angles to the surface.
- Example: water pressure acts perpendicular to the walls of a container and to a diver's body.

Q10. How can a small car lifter lift load heavier than itself?

Ans:

- A car lifter (hydraulic lift) works on Pascal's principle: pressure applied to an enclosed liquid is transmitted unchanged throughout it.
- A small force F_1 on the small piston (area A_1) produces pressure F_1/A_1, which acts on the large piston (area A_2).
- So the output force $F_2 = F_1 \times A_2/A_1$ is many times larger; the lifter acts as a force multiplier.